METRIC

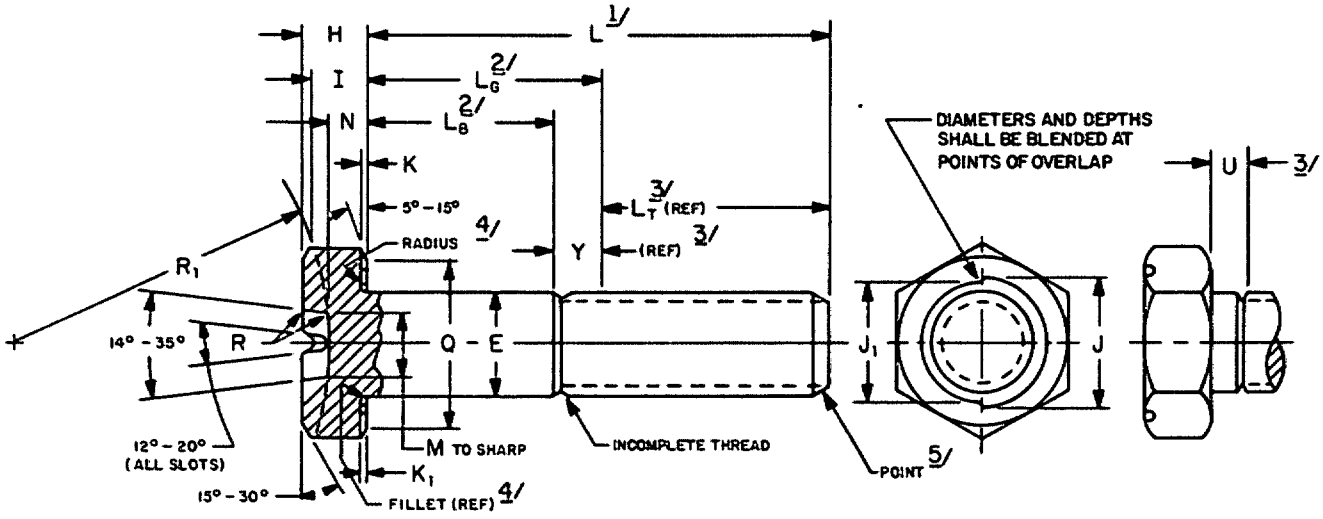
DOD-B-70331/2(AR)
21 April 1983

MILITARY SPECIFICATION SHEET

BOLT, SELF-LOCKING, HEXAGON HEAD, STEEL,
SAE J1199 (10.9) TYPE EH, METRIC

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the bolts described herein shall consist of this specification and the latest issue of DOD-B-70331.

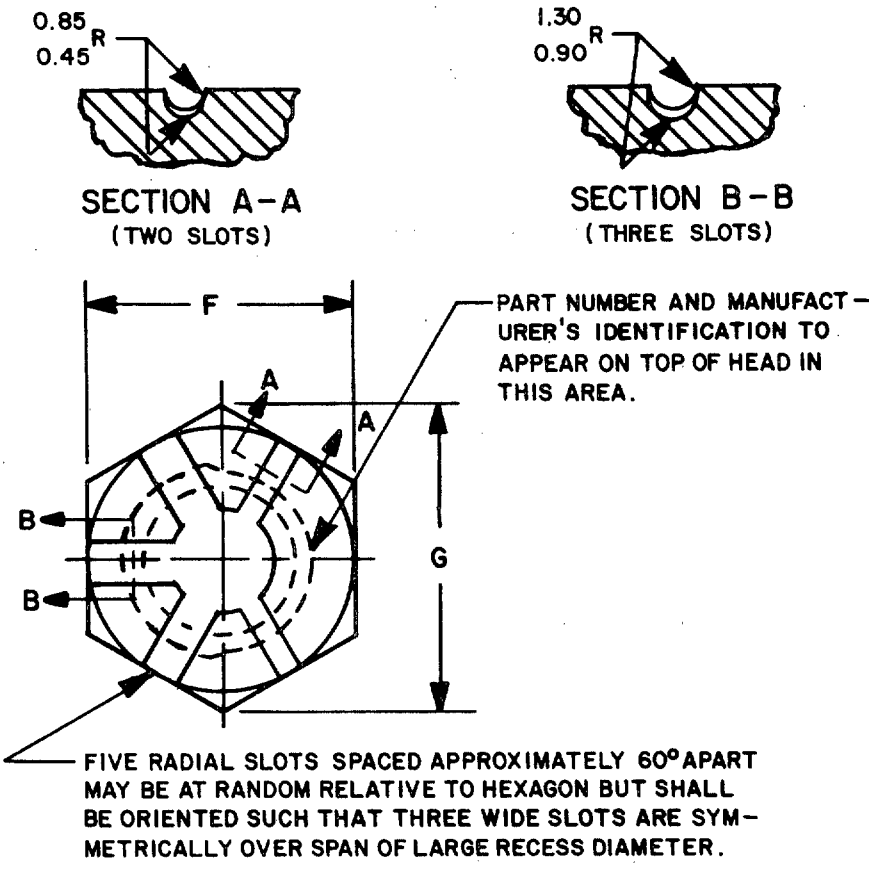


1/ See tables II and VI
2/ See table VI
3/ See table IV
4/ See table III
5/ See table V

FIGURE 1. Bolt, self-locking.

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FSC 5306



NOTES:

1. Bolt lengths above solid line in table VI are threaded to head.
2. All dimensions are in millimeters.
3. This bolt is covered by U.S. Patent Number 3342236, which expires 19 September 1984. The Government does not have a royalty free license under this patent. This bolt is known as a "place" bolt under this patent.

FIGURE 2. Bolt, self-locking, (detailed sections).

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TABLE I. Dimensions.

Dash no	Nom bolt diameter	E		F			G		H		I
		Body diameter		Hexagon width across flats			Hexagon width across corners		Head height		Wrenching height
		Max	Min	Basic	Max	Min	Max	Min	Max	Min	Min
6	M6X1	6.00	5.78	13	13.00	12.73	15.01	14.51	4.47	4.13	3.0
8	M8X1.25	8.00	7.78	15	15.00	14.70	17.32	16.76	5.50	5.10	3.7
10	M10X1.5	10.00	9.78	18	18.00	17.67	20.78	20.14	6.63	6.17	4.5
12	M12X1.75	12.00	11.73	21	21.00	20.64	24.25	23.53	7.76	7.24	5.2
14	M14X2	14.00	13.73	24	24.00	23.61	27.71	26.92	9.09	8.51	6.2
16	M16X2	16.00	15.73	27	27.00	26.16	31.18	29.82	10.32	9.68	7.0
20	M20X2.5	20.00	19.67	33	33.00	32.00	38.11	36.48	12.88	12.12	8.8
24	M24X3	24.00	23.67	40	40.00	39.00	46.19	44.46	15.44	14.56	10.5
30	M30X3.5	30.00	29.61	50	50.00	49.00	57.74	55.86	19.48	17.92	13.1

TABLE I. Dimensions. - Continued

Dash no	Nom bolt diameter	J		J_1		K		K_1	
		Large recess diameter		Small recess diameter		Large recess depth		Small recess depth	
		Max	Min	Max	Min	Max	Min	Max	Min
6	M6X1	8.29	8.04	7.80	7.55	0.17	0.10	0.17	0.07
8	M8X1.25	10.82	10.57	10.18	9.93	0.32	0.25	0.32	0.15
10	M10X1.5	12.98	12.73	12.21	11.96	0.32	0.25	0.32	0.15
12	M12X1.75	15.01	14.76	14.13	13.88	0.32	0.25	0.32	0.15
14	M14X2	17.54	17.29	16.51	16.26	0.32	0.25	0.32	0.15
16	M16X2	20.06	19.81	18.88	18.63	0.32	0.25	0.32	0.15
20	M20X2.5	24.94	24.69	23.48	23.23	0.32	0.25	0.32	0.15
24	M24X3	29.96	29.71	28.20	27.95	0.32	0.25	0.32	0.15
30	M30X3.5	37.82	37.57	35.60	35.35	0.32	0.25	0.32	0.15

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TABLE I. Dimensions. - Continued

Dash no	Nom bolt diameter	M		N		Q	R		R_1
		Head top recess diameter		Checking height		Bearing dia	Recess corner and fillet radius		Spherical radius
		Max	Min	Max	Min	Min	Max	Min	±6.35
6	M6X1	4.22	3.72	3.60	3.10	11.7	0.77	0.45	43.75
8	M8X1.25	5.29	4.79	4.37	3.87	13.5	0.77	0.45	43.75
10	M10X1.5	6.55	6.05	5.22	4.72	16.2	0.77	0.45	43.75
12	M12X1.75	7.81	7.31	6.07	5.57	18.9	0.77	0.45	43.75
14	M14X2	9.07	8.57	7.06	6.56	21.6	0.77	0.45	43.75
16	M16X2	10.33	9.83	7.99	7.49	24.3	0.77	0.45	43.75
20	M20X2.5	12.85	12.35	9.91	9.41	29.7	0.77	0.45	43.75
24	M24X3	15.37	14.87	11.83	11.33	36.0	0.77	0.45	43.75
30	M30X3.5	19.15	18.65	14.86	14.36	45.0	0.77	0.45	43.75

TABLE II. Length range tolerance.

Nominal bolt size range	M6X1 thru M8X1.25	M10X1.5 thru M16X2	M20X2.5 and M24X3	M30X3.5
Nominal bolt length	Length range tolerance, mm			
Up to 10 mm, incl	-0.6	-0.6	—	—
Over 10 to 18 mm, incl	-0.7	-0.7	—	—
Over 18 to 30 mm, incl	-0.8	-0.8	-1.3	—
Over 30 to 50 mm, incl	-1.0	-1.6	-2.5	-3.0
Over 50 to 80 mm, incl	-1.2	-1.9	-3.0	-3.5
Over 80 to 120 mm, incl	-1.4	-2.2	-3.5	-4.0
Over 120 to 180 mm, incl	-2.5	-2.5	-4.0	-4.5
Over 180 to 240 mm, incl	-4.6	-4.6	-4.6	-6.0
Over 240 mm	-6.0	-6.0	-6.0	-6.0

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TABLE III. Fillet dimensions.

Nominal bolt dia thread pitch	Fillet transition diameter		Fillet length		Fillet radius
	For short bolts	For short and long bolts	For long bolts	For short bolts	For short and long bolts
	Min	Max	Max	Max	Min
M6X1	6.2	6.8	1.1	0.7	0.3
M8X1.25	8.3	9.0	1.2	0.8	0.4
M10X1.5	10.2	11.0	1.2	0.9	0.4
M12X1.75	12.2	13.0	1.2	1.0	0.4
M14X2	14.1	15.0	1.2	1.1	0.4
M16X2	16.5	17.0	1.2	1.1	0.4
M20X2.5	20.7	21.0	1.2	1.2	0.4
M24X3	24.5	25.4	1.4	1.4	0.4
M30X3.5	30.7	31.4	1.6	1.6	0.4

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TABLE IV. Thread lengths.

Nom bolt dia thread size	L_T			U					Y
	Full form thread length (ref)			Unthreaded length under head					Transition thread length (ref)
	Basic								
	For bolt lengths			For bolt lengths	Max	For bolt lengths		Max	
	To 125 incl	Over 125 to 200 incl	Over 200	To and incl		Over	To and incl		Max
M6X1	18.0	24.0	37.0	12	1.50	12	25	2.50	5.0
M8X1.25	22.0	28.0	41.0	16	1.88	16	30	3.12	6.2
M10X1.5	26.0	32.0	45.0	20	2.25	20	35	3.75	7.5
M12X1.75	30.0	36.0	49.0	24	2.62	24	40	4.38	8.8
M14X2	34.0	40.0	53.0	28	3.00	28	45	5.00	10.0
M16X2	38.0	44.0	57.0	32	3.00	32	50	5.00	10.0
M20X2.5	46.0	52.0	65.0	40	3.75	40	60	6.25	12.5
M24X3	54.0	60.0	73.0	—	—	—	75	7.50	15.0
M30X3.5	66.0	72.0	85.0	—	—	—	90	8.75	17.5

TABLE V. Point diameter and length.

Nom bolt dia thread size	X	Z	
	Point diameter	Point length	
	Approx	Max	Min
M6X1	4.5	1.50	0.50
M8X1.25	6.1	1.88	0.62
M10X1.5	7.7	2.25	0.75
M12X1.75	9.4	2.62	0.88
M14X2	11.0	3.00	1.00
M16X2	13.0	3.00	1.00
M20X2.5	16.2	3.75	1.25
M24X3	19.5	4.50	1.50
M30X3.5	24.7	5.25	1.75

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TABLE VI. Maximum grip gaging lengths and minimum body lengths.

Dash no	L	Nom bolt dia									
		M6X1		M8X1.25		M10X1.5		M12X1.75		M14X2	
	Nom length	L_G Max	L_B Min	L_G Max	L_B Min	L_G Max	L_B Min	L_G Max	L_B Min	L_G Max	L_B Min
1	8										
2	10										
3	12										
4	14										
5	16										
6	20										
7	25										
8	30	12.0	7.0								
9	35	17.0	12.0	13.0	6.8						
10	40	17.0	12.0	13.0	6.8	14.0	6.5				
11	45	27.0	22.0	23.0	16.8	19.0	11.5	15.0	6.2		
12	50	27.0	22.0	23.0	16.8	19.0	11.5	15.0	6.2	16.0	6.0
13	55	37.0	32.0	33.0	26.8	29.0	21.5	25.0	16.2	21.0	11.0
14	60	37.0	32.0	33.0	26.8	29.0	21.5	25.0	16.2	21.0	11.0
15	65			43.0	36.8	39.0	31.5	35.0	26.2	31.0	21.0
16	70			43.0	36.8	39.0	31.5	35.0	26.2	31.0	21.0
17	75			53.0	46.8	49.0	41.5	45.0	36.2	41.0	31.0
18	80			53.0	46.8	49.0	41.5	45.0	36.2	41.0	31.0
19	85					59.0	51.5	55.0	46.2	51.0	41.0
20	90					59.0	51.5	55.0	46.2	51.0	41.0
21	100					74.0	66.5	70.0	61.2	66.0	56.0
22	110							80.0	71.2	76.0	66.0
23	120							90.0	81.2	86.0	76.0
24	130									90.0	80.0
25	140									100.0	90.0
26	150										
27	160										
28	170										
29	180										
30	190										
31	200										
32	220										
33	240										
34	260										
35	280										
36	300										

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TABLE VI. Maximum grip gaging lengths and minimum body lengths. - Continued

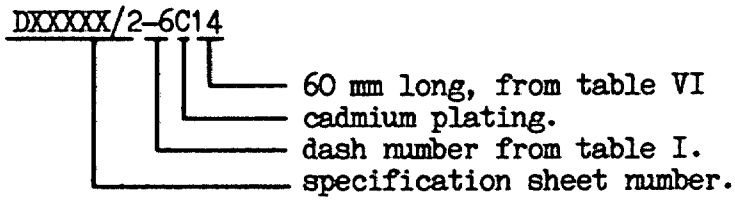
Dash no	L	Nom bolt dia							
		M16X2		M20X2.5		M24X3		M30X3.5	
	Nom length	L_G Max	L_B Min	L_G Max	L_B Min	L_G Max	L_B Min	L_G Max	L_B Min
1	8								
2	10								
3	12								
4	14								
5	16								
6	20								
7	25								
8	30								
9	35								
10	40								
11	45								
12	50								
13	55	17.0	7.0						
14	60	17.0	7.0						
15	65	27.0	17.0	19.0	6.5				
16	70	27.0	17.0	19.0	6.5				
17	75	37.0	27.0	29.0	16.5				
18	80	37.0	27.0	29.0	16.5	26.0	11.0		
19	85	47.0	37.0	39.0	26.5	31.0	16.0		
20	90	47.0	37.0	39.0	26.5	36.0	21.0		
21	100	62.0	52.0	54.0	41.5	46.0	31.0	34.0	16.5
22	110	72.0	62.0	64.0	51.5	56.0	41.0	44.0	26.5
23	120	82.0	72.0	74.0	61.5	66.0	51.0	54.0	36.5
24	130	86.0	76.0	78.0	65.5	70.0	55.0	58.0	40.5
25	140	96.0	86.0	88.0	75.5	80.0	65.0	68.0	50.5
26	150	106.0	96.0	98.0	85.5	90.0	75.0	78.0	60.5
27	160	116.0	106.0	108.0	95.5	100.0	85.0	88.0	70.5
28	170			118.0	105.5	110.0	95.0	98.0	80.5
29	180			128.0	115.5	120.0	105.0	108.0	90.5
30	190			138.0	125.5	130.0	115.0	118.0	100.5
31	200			148.0	135.5	140.0	125.0	128.0	110.5
32	220					147.0	132.0	135.0	117.5
33	240					167.0	152.0	155.0	137.5
34	260							175.0	157.5
35	280							195.0	177.5
36	300								

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Requirements:

1. Part number. The part number consists of the letter "D", the basic number of this specification sheet, followed by a dash number from table I. Add "C" for cadmium plating or "P" for phosphate coating followed by a dash number from table VI.

Example:



Custodians:
Army - AR

Review activities:
Army - MI
DLA - IS
NSA - NS

User activities:
Army - AT, GL, ME
Navy - MC, OS, SH,

Agent:
DLA - IS

Preparing activity:
Army - AR

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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